

Wi-Fi CAN TRACK YOUR HEART RATE Without Any Wearables

No wristbands, no patches, no gadgets—just Wi-Fi. Find out how your heartbeat can be tracked invisibly using signals already in your home.



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In recent years, growing public interest in health has made ubiquitous monitoring a key focus for researchers and users alike. Traditional methods for tracking vital signs rely on wearables that make contact with the body. Although these devices offer reasonable accuracy, they can be inconvenient for long-term use and are not always practical in daily routines. It is now possible, however, to measure heart rate without any physical contact, not even a lightweight wearable.

Imagine sitting in your drawing room while your heart rate is monitored automatically. This is now feasible using Wi-Fi signals already present in homes or offices. When these signals pass through the body, they undergo small changes. By analysing subtle variations in Channel State Information (CSI), the system can detect shifts in heart rate without touch. This non-contact approach improves comfort and increases real-world usability by

eliminating the need for continuous user input.

Its appeal lies in its low cost, convenience, and unobtrusiveness. Earlier studies could detect breathing using Wi-Fi but struggled to track heartbeats because the signal changes were extremely faint. By combining CSI with machine learning (ML), current methods filter out irrelevant noise, including reflections from stationary objects, and isolate patterns linked to different heart rates.

How signals reveal your pulse

As mentioned earlier, in Wi-Fi heart-rate monitoring, CSI is used to analyse how chest movements from the heartbeat affect the Wi-Fi signal. The system captures fine-grained CSI data from standard Wi-Fi devices and uses signal processing and machine learning to extract subtle changes caused by the pulse. Typically, the heartbeat frequency is estimated from peaks in the signal.